When Alzheimer's EEG Does Not Look Old: Brain-Age Gap and CWT-Scalogram CNNs for Interpretable Alzheimer's Disease Detection

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Abstract—Resting-state electroencephalography (EEG) is a low-cost candidate for Alzheimer's disease (AD) screening. We present an interpretable EEG pipeline with two complementary modules, evaluated on public data with strictly subject-independent validation. The first is a brain-age model trained only on healthy adults. It was trained on 200 adults aged 20–70 years and transferred to a clinical cohort of 36 AD patients, 29 healthy controls (HC) and 22 frontotemporal dementia (FTD) patients. The second is a compact convolutional neural network (CNN) on continuous-wavelet-transform (CWT) scalograms, explained with Grad-CAM. Contrary to the accelerated-aging hypothesis, the bias-corrected brain-age gap (BAG) was strongly negative in AD (-11.2 ± 14.2 vs. 0.0 ± 8.7 years in HC; Cohen's $d = -0.93$, 95% CI $[-1.52, -0.46]$). This effect was specific to AD (FTD vs. HC: $d = -0.17$). It held across all numerically stable model configurations, on ICA-cleaned data, and in a replication with 513 healthy training subjects. Feature-level analysis explains the reversal: slow-wave power falls and beta power rises with healthy aging, whereas AD shows the opposite. The CNN classified AD vs. HC with AUC 0.89 and accuracy 82% (mean of three seeds). This was statistically indistinguishable from spectral-feature baselines and EEGNet. Adding the BAG did not improve it (ΔAUC $= 0.01$, p $= 0.74$). Grad-CAM attributed AD decisions mainly to theta and alpha activity. We conclude that EEG brain age in AD reflects a deviation from normative aging rather than accelerated aging.

Index Terms—Alzheimer's disease, EEG, brain age, continuous wavelet transform, convolutional neural network, Grad-CAM, explainable AI

I. Introduction

Alzheimer's disease is the most common cause of dementia. Diagnosis currently relies on clinical assessment, neuropsychological testing and costly imaging such as MRI and PET, which are scarce in resource-limited settings. Resting-state EEG is inexpensive, non-invasive and widely available, and AD alters it in a characteristic way: oscillatory "slowing", with more delta and theta and less alpha and beta activity [1], [2].

Two families of computational methods exploit EEG for AD. Discriminative classifiers learn to separate patients from controls directly, using hand-crafted spectral features or deep networks on raw or time–frequency representations [3], [4]. Brain-age models are trained only on healthy people to predict chronological age. The brain-age gap (BAG, predicted minus chronological age) is then read as a marker of accelerated aging and has been used widely in neuroimaging [5], [6]. Classifiers often achieve high accuracy but are opaque. Brain age offers an intuitive, continuous score, but its assumption that disease "ages" the brain is rarely tested for EEG.

This work combines both approaches in one reproducible pipeline and tests them under strict subject-independent evaluation. Our contributions are:

1) A harmonized EEG brain-age module trained on healthy adults and transferred to a clinical AD cohort without ever training on patients. We show that the AD brain-age gap is robustly negative and AD-specific, and we explain this through opposite spectral changes in aging and AD.
2) A compact CWT-scalogram CNN with Grad-CAM explanations, compared on identical subject folds with four baselines, with ablations and a three-class (HC/AD/FTD) extension.
3) An honest assessment of combining the two modules. The BAG adds interpretability, not accuracy.
4) An open, end-to-end implementation, including an interactive application that produces a per-person report.

II. Related Work

A. EEG markers of Alzheimer's disease

Reviews of EEG in AD consistently report slowing of the dominant rhythm, increased delta and theta power, and decreased alpha and beta power. These changes are often accompanied by reduced synchronization [1], [2]. Babiloni et al. emphasize that these changes differ from those of normal aging [2]. This distinction is central to our brain-age findings.

B. Machine learning and deep learning on EEG

Classical pipelines extract band powers or spectral ratios and apply classifiers such as SVMs or random forests. Deep learning removes manual feature design: CNNs on raw signals or on time–frequency images such as spectrograms and wavelet scalograms have been applied to many EEG tasks [3], [4]. EEGNet is a widely used compact architecture for raw EEG [7]. Reviews caution that small datasets and subject-dependent splits often

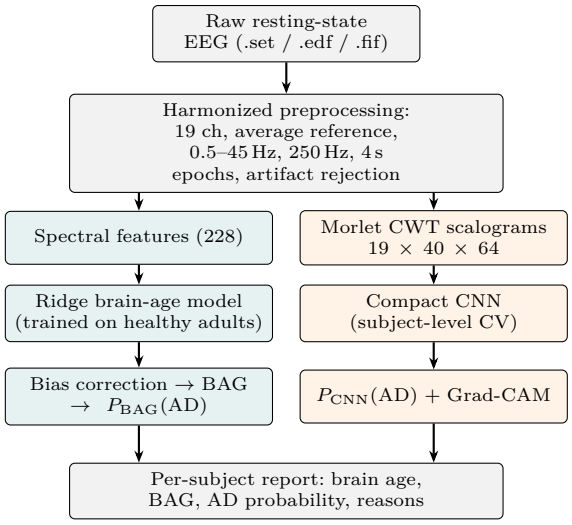


Fig. 1. Proposed pipeline. Teal: brain-age module. Orange: CNN classifier.

inflate reported accuracy [4], [8]. For this reason we use subject-level splits and strong baselines throughout.

C. Brain age

Brain-age prediction from MRI is an established biomarker framework [5], [6]. Statistical bias toward the mean age must be corrected [9]. EEG-based brain age has been estimated from resting-state recordings [10], [11]. Sleep-EEG brain age was elevated in people with dementia [12], supporting the accelerated-aging view. Whether this holds for resting-state EEG in clinically diagnosed AD, using a model that never sees patients, has not been established.

D. Explainability and harmonization

Grad-CAM produces class-specific importance maps from the last convolutional layer of a CNN [13]. On scalograms these maps can be read as frequency-by-time importance. Combining data from different sites requires harmonization; ComBat [14] removes site effects while protecting biological covariates and has been adopted for neuroimaging [15].

III. Proposed Methodology

Fig. 1 summarizes the system. One harmonized preprocessing stage feeds two modules: a brain-age module trained on healthy adults, and a CWT-scalogram CNN classifier. Their outputs are reported side by side and can be combined.

A. Harmonized preprocessing

A single function processes training data, test data and new uploads, using MNE-Python [16]. It selects 19 channels in a fixed order (10–10 names mapped to 10–20), applies an average reference over these channels, a 0.5–45 Hz zero-phase FIR band-pass and resampling to 250 Hz. It then cuts 4 s epochs with 2 s overlap and rejects epochs whose peak-to-peak amplitude exceeds 150 μV or falls below 1 μV. ICA is not applied, so the pipeline is identical across datasets and deployment. The effect of this choice is tested with the dataset authors' ICA-cleaned data.

B. Brain-age module

For each subject, the Welch power spectrum (4 s Hann window) is averaged over epochs. From it we compute per channel: log absolute and relative power in the delta (1–4 Hz), theta (4–8), alpha (8–13) and beta (13–30 Hz) bands; the alpha peak frequency; the log theta/alpha ratio; and the aperiodic exponent and offset (FOOOF [17]). This gives 228 features. A ridge regressor predicts age. Because the clinical cohort spans a narrow age range, ds004504 HC are added to the healthy training set, and every HC prediction is made out-of-fold. Age bias is corrected on HC predictions [9], and the BAG is mapped to $P(\mathrm{AD})$ with logistic regression. AD and FTD subjects never enter training, harmonization or model selection.

C. CWT-scalogram CNN

Each epoch becomes a Morlet scalogram with 40 log-spaced frequencies (1–40 Hz) and 64 time bins, a $19 \times 40 \times 64$ tensor with EEG channels as input feature maps. The CNN has three blocks of convolution (3×3), batch normalization and ReLU (16, 32 and 32 filters, the first two followed by max-pooling), then global average pooling, dropout of 0.5 and a linear output. It has about 17k parameters, deliberately small for a 65-subject cohort. A subject's probability is the mean over its epochs. Grad-CAM [13] on the last block yields frequency-by-time importance.

D. Combination and evaluation protocol

The two modules are combined by stacking: logistic regression on [logit P_{CNN}, BAG]. All evaluation is subject-independent. The CNN uses subject-level 10-fold cross-validation repeated with three seeds, each changing both the fold split and the initialization. Baselines use identical folds. We report AUC, accuracy, precision, recall, specificity and F1, with 95% bootstrap CIs (2000 subject resamples). AUC significance is assessed by permutation tests that re-run the entire CV on shuffled labels [18], and model differences by a paired bootstrap.

IV. Algorithms

Algorithms 1–4 specify the pipeline. Algorithm 2 guarantees that no patient influences the age model or its bias correction. Algorithm 3 guarantees that normalization, early stopping and testing never share subjects. Algorithm 4 runs in about 10 s per recording on a laptop CPU.

Algorithm 1 Harmonized preprocessing

Require: EEG recording R; channel list $\mathcal{C}$ (19, 10–20)
Ensure: clean epochs $X \in \mathbb{R}^{n \times 19 \times 1000}$

map 10–10 names (T7, T8, P7, P8) to 10–20 (T3, T4, T5, T6)
if any channel of $\mathcal{C}$ is missing or duration < 30 s then
    reject with an explanatory message
end if
select $\mathcal{C}$ in fixed order; re-reference to their average
band-pass 0.5–45 Hz (zero-phase FIR); resample to 250 Hz
cut 4 s epochs with 2 s overlap
keep epoch e iff $1\,\mu\text{V} < \text{ptp}_c(e) < 150\,\mu\text{V}$ for all channels c
if fewer than 10 epochs remain then
    reject
end if
return X (cached by configuration hash)

Algorithm 2 Brain-age training, transfer and BAG

Require: healthy cohort D_H (age 20–70); target HC set T_C; patients T_P (AD, FTD)
Ensure: BAG for every target subject

for each $s \in T_C$ do ▷ leave-one-HC-out
    fit ridge (inner-CV α) on features of $D_H \cup T_C \setminus \{s\}$
    $\hat{y}_s \leftarrow$ predicted age of s
end for
fit ridge on $D_H \cup T_C$; $\hat{y}_p \leftarrow$ prediction for each $p \in T_P$
fit $\hat{y} = a\,y + b$ on $\{(y_s, \hat{y}_s) : s \in T_C\}$ ▷ HC only
$\text{BAG}_i \leftarrow (\hat{y}_i - b)/a - y_i$ for all target subjects i
return BAG; fit $P(\text{AD} \mid \text{BAG})$ by logistic regression

Algorithm 3 Subject-level cross-validation of the CNN

Require: subjects S with labels; scalograms Z_s (≤ 150 per subject); seed r
Ensure: out-of-fold $P(\text{AD})$ for every subject

split S into 10 stratified folds $F_1, \ldots, F_{10}$ (seed r)
for $k = 1$ to 10 do
    $S_{\text{tr}} \leftarrow S \setminus F_k$; split S_{tr} into fit (85%) and validation (15%) subjects
    compute mean/SD per channel×frequency on fit subjects only
    train CNN on fit epochs (Adam, weighted BCE)
    stop when validation-subject loss has not improved for 6 epochs; restore best weights
    for each $s \in F_k$ do
        $P_s \leftarrow$ mean of epoch probabilities of s
    end for
end for
return $\{P_s\}$ ▷ no subject is in both training and test

Algorithm 4 Inference for a new subject

Require: EEG file, chronological age y

$X \leftarrow$ Algorithm 1(file)
brain age $\hat{y} \leftarrow$ ridge(features(X)); $\text{BAG} \leftarrow (\hat{y}-b)/a-y$
$P_{\text{CNN}} \leftarrow$ mean CNN probability over scalograms of X
Grad-CAM band importance toward the predicted class
compare alpha peak, band ratios and BAG with HC/AD reference values
return report: brain age, BAG, P_{CNN}, P_{BAG}, combined P, reasons

V. Dataset

Clinical cohort (OpenNeuro ds004504) [19]. Resting-state eyes-closed EEG from 36 AD, 29 HC and 22 FTD subjects aged roughly 44–79, recorded clinically with 19 electrodes (10–20 system) at 500 Hz. Labels include MMSE scores. One FTD recording failed artifact screening. The authors also provide ICA-cleaned derivatives, which we used only for sensitivity analysis. The dataset contains no mild-cognitive-impairment subjects, so the classification task is AD vs. HC, with FTD as an additional class.

Healthy aging cohort (OpenNeuro ds005385, Dortmund Vital Study) [20]. 64-channel EEG at 1000 Hz from healthy adults aged 20–70. We used the baseline session, eyes-closed, pre-task recordings. Recordings flagged as possibly discontinuous were excluded. The main analysis used an age-stratified sample weighted toward older ages ($n = 200$; 25/25/35/50/65 per decade); a replication used all 517 eligible subjects (513 after preprocessing).

After preprocessing, the clinical subjects kept on average about 80% of their epochs, and the healthy cohort about 90%. The median peak-to-peak amplitude was similar in the two cohorts, which confirms consistent units and scaling.

VI. Results

A. Brain age: AD departs from the healthy-aging pattern

On the healthy cohort the brain-age model reached MAE 9.2 years ($r = 0.60$) in 10-fold CV. On ds004504 HC the out-of-fold MAE was 7.4 years ($r = 0.53$, bias slope $a = 0.88$). The out-of-fold R^2 was -1.9, which is expected for an HC age range of only 57–78 years, so R^2 is not used as the success criterion. The bias-corrected BAG was 0.0 ± 8.7 years in HC, -11.2 ± 14.2 in AD and -2.2 ± 17.8 in FTD (Fig. 2). AD differed from HC (Welch $t = -3.90$, $p < 0.001$; Mann–Whitney $p < 0.001$; $d = -0.93$, 95% CI $[-1.52, -0.46]$). FTD did not differ from HC ($d = -0.17$, $p = 0.87$), and AD differed from FTD ($d = -0.57$, $p = 0.04$). The negative AD gap appeared in 12 of 12 numerically stable configurations (d from -0.99 to -0.26). It persisted on ICA-cleaned data ($d = -0.70$, $p = 0.02$) and in the 513-subject replication ($d = -0.83$, permutation $p = 0.003$). Within AD, BAG was not associated with MMSE ($\rho = -0.08$, $p = 0.62$).

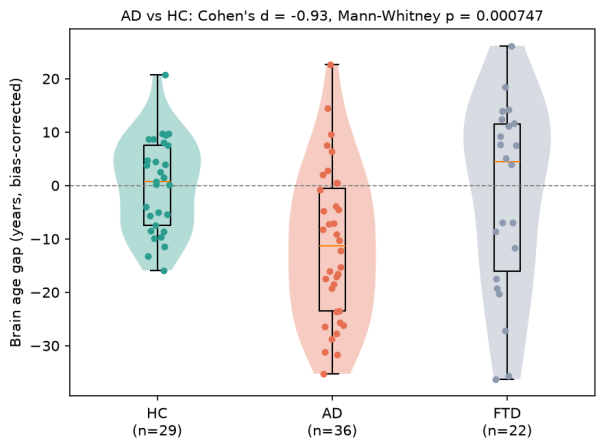


Fig. 2. Bias-corrected brain-age gap by group. HC are out-of-fold; AD and FTD were never used for training.

TABLE I
Healthy-aging direction vs. AD effect per feature family

Feature	r (age, healthy)	d (AD−HC)	AD looks
Alpha peak freq	−0.25	−1.47	older
Rel. alpha	−0.30	−1.33	older
Abs. alpha	−0.32	−0.96	older
Theta/alpha ratio	+0.18	+1.75	older
Rel. theta	−0.05	+1.55	younger
Abs. theta	−0.29	+1.03	younger
Rel. delta	+0.11	+1.15	older
Abs. delta	−0.26	+1.04	younger
Rel. beta	+0.35	−0.99	younger
Abs. beta	−0.06	−0.80	older
Aperiodic exponent	−0.28	+0.95	younger
Aperiodic offset	−0.38	+0.98	younger

Table I explains the reversal. Alpha peak frequency, alpha power and the theta/alpha ratio change in the same direction in aging and AD. In contrast, beta power rises with age but falls in AD, while absolute slow-wave power and the aperiodic exponent fall with age but rise in AD. A model of healthy aging therefore places AD EEG off, and on balance "before", the normative trajectory. Used alone as a classifier, BAG reached AUC 0.70 (95% CI [0.56, 0.82], permutation p = 0.002), accuracy 66%, precision 68%, recall 72% and specificity 59%, with the decision rule BAG < -2.6 years.

B. CNN classification and baselines

Over three seeds the CNN achieved AUC 0.89 (range 0.86–0.90), accuracy 82%, precision 84%, recall 83%, specificity 80% and F1 0.84. For the primary seed the AUC was 0.86 (95% CI [0.76, 0.94]) and the accuracy 80% (95% CI [69, 89]%). In a permutation test with $N = 72$ full re-runs on shuffled labels, the null AUC averaged 0.50 (maximum 0.72), giving p = 0.01. On identical folds (Table II) all models performed comparably (AUC 0.84–0.88; highest: EEGNet (raw EEG), 0.88). No difference from the CNN was significant (all $p \geq 0.63$). With $n = 65$, models within about 0.1 AUC cannot be distinguished [8].

C. Combining the two modules

Stacking the CNN with BAG gave AUC 0.87 versus 0.86 for the CNN alone ($\Delta = 0.01$, 95% CI $[-0.06, 0.08]$, p = 0.74). Both modules draw on the same EEG slowing, so the BAG adds interpretability rather than accuracy.

D. Ablations

Table III varies one design choice at a time (three seeds each). No change reduced the mean AUC by more than 0.05, which is comparable to the variation between seeds. Using only 8–40 Hz (AUC 0.87) kept more information than using only 1–8 Hz (0.85), and collapsing the time axis (0.84) gave a small drop. Eight channels (0.85) and 40% of the epochs (0.88) stayed close to the full model (0.89). This suggests that low-density montages and shorter recordings are viable.

E. Three-class classification

For HC/AD/FTD the CNN reached balanced accuracy 63% (macro AUC 0.78), versus 56% (0.72) for a random forest on spectral features. Class recall was 76% for HC, 69% for AD and 45% for FTD; 10 of 22 FTD subjects were classified as AD (Table IV). This is consistent with the overlapping EEG slowing of the two dementias.

F. Explainability

Relative to the average over frequencies (1.0), final-model Grad-CAM importance for AD decisions was theta 1.29, alpha 1.39, delta 0.83 and beta 1.07. For HC decisions it was alpha 1.37 and beta 1.40 (Fig. 3). Out-of-fold Grad-CAM, from fold models that never saw the explained subject, reproduced this pattern: AD theta 1.29 and alpha 1.27, and HC alpha 1.35 and beta 1.47. The most important channels for AD were temporo-parietal (F8, T6, T3, P4). This agrees with the known theta increase and alpha decrease in AD.

VII. Conclusion and Contribution

This work tested two interpretable EEG approaches to Alzheimer's disease under strictly subject-independent evaluation on public data. Its contributions and conclusions are:

1) The brain-age gap in AD is a deviation, not acceleration. A brain-age model that never saw patients found AD EEG younger, not older ($d = -0.93$). The effect was AD-specific, robust across configurations, preprocessing and training sets, and explained by spectral features that change in opposite directions in aging and AD. EEG brain age in AD should therefore be interpreted as a normative-deviation score.

2) A compact, explainable CNN. The CWT-scalogram CNN reached AUC 0.89, above the 0.75 target, with Grad-CAM explanations centred on theta and alpha. It was statistically indistinguishable from spectral baselines

TABLE II
AD vs. HC on identical subject-level 10-fold splits (primary seed, $n = 65$). Last column: CNN AUC minus model AUC (paired bootstrap).

Model	AUC	Acc. (%)	Recall (%)	Spec. (%)	F1	ΔAUC (CNN−model)
Spectral features + logistic regression	0.84	75	72	79	0.76	+0.02 (p = 0.68)
Spectral features + RBF-SVM	0.85	80	75	86	0.81	+0.01 (p = 0.89)
Spectral features + random forest	0.86	77	75	79	0.78	+0.00 (p = 0.94)
EEGNet (raw EEG)	0.88	80	81	79	0.82	−0.02 (p = 0.63)
CWT-scalogram CNN (ours)	0.86	80	83	76	0.82	–

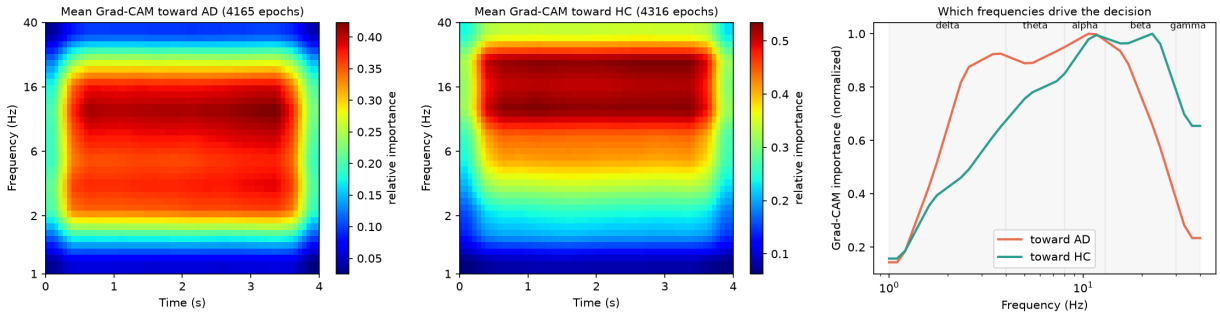


Fig. 3. Grad-CAM of the final CNN: mean maps toward AD and HC, and frequency profiles.

TABLE III
CNN ablations (mean ± SD over seeds)

Variant	AUC	Acc. (%)	seeds
Full model (19 ch, 1–40 Hz, time)	0.89 ± 0.03	82 ± 2	3
Only 1–8 Hz	0.85 ± 0.01	77 ± 4	3
Only 8–40 Hz	0.87 ± 0.02	79 ± 2	3
No time axis (spectrum)	0.84 ± 0.04	79 ± 5	3
8 channels	0.85 ± 0.02	79 ± 3	3
40% of epochs	0.88 ± 0.03	78 ± 2	3

TABLE IV
Three-class CNN confusion matrix (primary seed; rows: true class)

True \ predicted	HC	AD	FTD
HC	22	3	4
AD	6	25	5
FTD	2	10	10

and EEGNet, which is a realistic picture for small EEG cohorts.

3) The value of combining the modules lies in interpretability. The BAG did not improve classification, but it provides a continuous, explainable score. Both are delivered in a per-subject report by an open application.

Limitations and future work. The cohort is small ($n = 65$ for AD vs. HC) and there is no external test set. The healthy cohort ends at 70 years, while the clinical HC reach 78. ICA was not used in the main pipeline. BAG did not track MMSE, and the data contain no MCI subjects, so progression could not be studied. Future work will validate the pipeline on independent cohorts, extend it to MCI and longitudinal data, and explore normative modeling of EEG spectra.

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